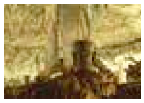

Muon maps of pyramids
Researchers are using advances in high energy physics to probe the pyramids for hidden chambers. **https://dgit.in/jun20-66**


Mayan sacred cave
How 3D laser scanning helped visualise a hidden sacred cave belonging to the Mayans. **https://dgit.in/jun20-67**

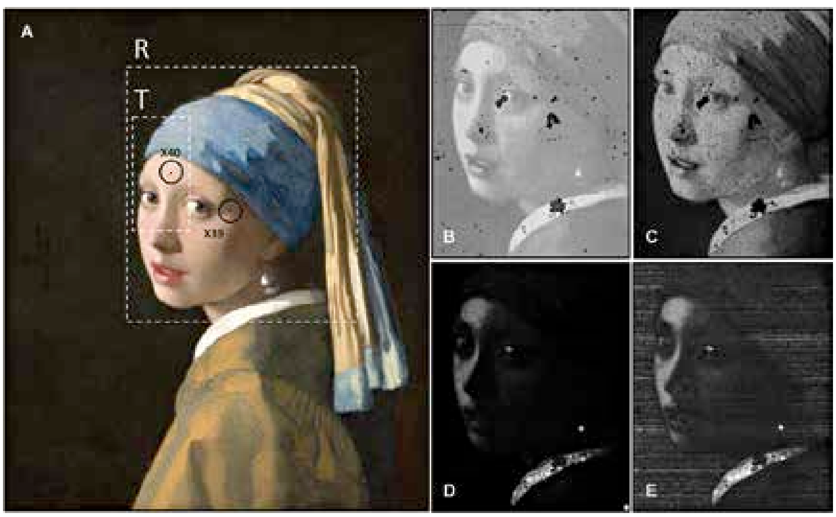
up and sealed within cylinders – exposing them may destroy them. From ancient Israel, there are shards of ceramic, called ostraca, that were used as ancient versions of post-it notes that are studied using multi-spectral imaging. On a well studied piece of ostraca which had a blessing in the front, researchers found another message on the back which was previously believed to be blank. The message called for more wine, and promised anything the sender wanted in return.

Hyperspectral imaging has also been used to study the writings on cartonnage, the papyrus that is used to wrap mummies. Remains at Herculaneum, a Roman resort that was buried in the eruption of Vesuvius has a well preserved library of scrolls recovered from the house of a statesman. Most of the scrolls have never been opened, as the few that have crumbled, revealing only fragments of text. The collection is the biggest hoard of text from the time period. Previous archaeologists would cut through the scrolls, open them up, copy the text, scrape away a layer, and then copy the text again. As the scroll was deciphered, it would also get destroyed. Now, the new technique allows researchers to look inside the scrolls and read the letters without destroying them. The approach has also been used on dead sea scrolls.

PXRF

X-ray fluorescence is a process that allows researchers to analyse the elemental composition of samples by bombarding an object or area with X-rays. It also provides the density of a certain element. In function, the device is similar to the Tricoder from Star Trek. The particles of the material that are excited by these X-Rays then give out secondary X-rays, a phenomenon known as fluorescence. Traditionally, these were large machines into which samples had to be loaded. Now, miniaturised versions can be taken into the field, and these portable X-Ray Fluorescence (pXRF) allows archaeologists to study objects non-destructively. This can be useful for exploring finds that would have been otherwise impossible to detect, such as rock paintings that have subsequently been covered by more rocky material. Digging it up would just destroy the thin layer of pigments. The device was used to explore the massive 2000-year-old rock paintings in the Lower Pecos Canyonlands in Texas. The archaeologists were also able to determine that charcoal was used as a pigment in painted pebbles found in the area. As the device can provide information on the chemical properties and compositions of the objects being studied, it can be used to find out extremely specific things about more recent artefacts. A study on Vermeer's Girl with Pearl Earring showed that the artist was very particular about which paints he used where, and that he had used four different types of white paint in the painting. In the 1980s, Indonesian fishermen came across a shipwreck, loaded with pottery from China. The 800-year-old ceramic pots were made from a bluish white porcelain known as qinbai. Researchers had a hunch that the pottery were made in the kiln complexes at Jingdezhen, Dehua, Huajiashan, and Minqing in China, but this guess was not confirmed, and other kiln complexes could not be ruled out. A team of researchers decided to settle the matter by using pXRF, and matching the elemental composition of the pot with shards found at the site of the kiln complexes. They were able to confirm the guess. Finding out the origin of objects is really helpful for archaeologists as it allows them to find the dates of finds without any biological material on them, where radiocarbon dating is not possible. For example, objects made of copper from Europe found in the Americas can only be so old.

Faigenbaum-Golovin, S. et al./PLOS ONE



Four different white pigments used in Johannes Vermeer's Girl with Pearl Earring, revealed through XRF.